Analyzing Pedagogical Quality and Efficiency of LLM Responses with TA Feedback to Live Student Questions

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Abstract

While Large Language Models (LLMs) have emerged as promising methods for automated student question-answering, guaranteeing consistent instructional effectiveness of the response remains a key challenge. Therefore, there is a need for fine-grained analysis of State-Of-The-Art (SOTA) LLM-powered educational assistants.

This work evaluates Edison: a Retrieval Augmented Generation (RAG) pipeline based on GPT-4. We determine the pedagogical effectiveness of Edison's responses through expert Teaching Assistant (TA) evaluation of the answers. After the TA edits and improves the response, we analyze the original LLM response, the TA-assigned ratings, and the TA's edits to ascertain the essential characteristics of a high-quality response. Some key insights of our evaluation are as follows: (1) Edison can give relevant and factual answers in an educational style for conceptual and assignment questions, (2) Most TA edits are deletions made to improve the style of the response, and finally (3) Our analysis indicates that Edison improves TAs' efficiency by reducing the effort required to respond to student questions.

CCS Concepts

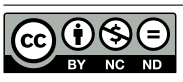
• **Applied computing** → **Computer-assisted instruction**; • **Computing methodologies** → **Natural language generation**; **Discourse, dialogue and pragmatics**.

Keywords

Language models, Question answering, Instructional technologies, Expert feedback

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1 Introduction

Interest in Computer Science (CS) has grown at a rapid rate [3], driving automated question-answering with Large Language Models (LLMs) [1, 11] due to their demonstrated strong performance in answering subject-specific questions [14] and generating code [5]. Despite the increased use of LLM-based assistants for fast and personalized responses to student questions, challenges remain in evaluating the instructional effectiveness of LLM-generated responses, especially for course-specific questions.

When evaluating LLM tools [25, 34, 43] for coding assistance or answering questions in CS courses, the key question is how to evaluate the instructional effectiveness of responses. Beyond metrics like relevance and factuality [17, 25, 31], recent work highlights the importance of interactivity, time-saved, and withholding of direct solutions in automated responses [8, 18, 43] as key factors impacting response quality in educational settings. Most works address this by using statistical analysis, student and educator surveys, and human evaluations [8, 18, 41].

Our work evaluates the pedagogical quality and efficiency of LLM responses through expert TA feedback. To achieve this, we first present our LLM assistant, Edison, a GPT-4-based method that leverages Optical Character Recognition (OCR), retrieval-augmented generation (RAG)[23, 35], and prompt engineering (including expert pre-conditioning and few-shot examples) to answer course-specific natural language and coding questions consisting of text and/or images. We deploy Edison with TAs in the loop in a large-scale upper-division undergraduate data science course, Data 100, at UC Berkeley.

Our evaluation uses expert TA feedback to assess the relevance, factuality, and style of Edison responses. We analyze TA edits to Edison responses to answer the following research questions:

RQ1: How do TAs evaluate Edison's responses on factuality, relevance, and style?
RQ2: What are the main attributes of TAs' edits to Edison's responses?
RQ3: What are the determining factors of Edison's responses that lead TAs to completely rewrite the answers from scratch?
RQ4: Are there efficiency gains for TAs from the use of Edison?

Our analysis shows that Edison gives relevant and factual responses, struggling most with style due to response verbosity. As a result, most TA edits of Edison responses are deletions. Additions made by TAs focused on making the responses more relevant and factual. Overall, we hope that this detailed analysis of the strengths and weaknesses of an LLM-based educational assistant can guide future work in the development and evaluation of these systems.

2 Related Work

2.1 LLM-powered Virtual Assistants in Education

Logistical inefficiencies and repeated questions are major challenges of scaling office hours for large classes [27]. A study on CS instructor pain points found that much time is spent assisting students with individual or group queries [29], and Oberman notes a link between student motivation and help-seeking in CS courses [6], highlighting the value of digital tools for reducing help-seeking barriers [2, 4]. To address these challenges, research has focused on the adoption of LLMs such as GPT-4 [1], LLaMA [10], and Gemini [11] as educational assistants [12, 18, 25, 31, 42, 43].

One LLM application is answering student questions [15, 31], including use in discussion forums and virtual office hours. Other educational applications of LLMs include code generation [36, 38, 40], student submission feedback [17, 21, 28], student modeling [9, 26], and course content development [33, 37, 39].

2.2 Pedagogical Evaluation of LLMs

The continual improvement and adaptation of intelligent tutor systems via data-driven approaches also guides LLM educational assistant development [19, 32]. While many benchmarks exist for general LLM tasks like question-answering [14] and conversation [7], educational assistants require higher standards for evaluating factuality, relevance, and response style. Automated metrics like ROGUE [24], and similarity-based embedding methods [35] provide quick insight into the similarity of LLM-generated and expert responses but lack depth in assessing factuality or pedagogical style and often overfit to the structure and template of the text rather than the content. As such, these methods struggle to evaluate syntactically similar but semantically or structurally diverse answers to questions [16].

The best existing method for pedagogical evaluation is to assess LLM assistants on real student questions from courses [18, 31, 41, 43], leveraging human evaluation [13, 18, 25, 30, 31], surveys [18, 25], and interviews [8, 13, 18] from students and educators regarding LLM assistant interactions. These methods allow for fine-grained analysis of instructional effectiveness.

3 Method

3.1 Deployment Settings

Our team deployed Edison in a large-scale upper-division undergraduate data science course, Data 100[1], at the University of California, Berkeley, during the Spring 2024 semester. The course served 1133 undergraduate and 71 graduate students and was operated by 59 staff members. The course was chosen in part due to the diversity of topics discussed (data processing, machine learning, probabilistic modeling, etc.) and the different types of questions that arise as a consequence (logistics, mathematics, programming, conceptual, etc.)

Edison was integrated into the course discussion forum — EdSTEM[2] — for one month[3]. During this timeframe, for each new question, students were asked to provide consent if they permitted the use of Edison. Upon consent, a TA would determine whether the question was appropriate to send to Edison; factors for this determination included the presence of personally identifiable information and extension requests. If determined to be appropriate, the TA triggered Edison, which returned a TA-visible response to the student question. The TA was subsequently prompted to rate the generated response on a scale from 1 to 5 for the following three metrics — factuality, relevance, and style (see metric definitions in Table 1). We measure the pedagogical quality of a response as a combination of these three metrics. The TA then decided whether to directly use, edit, or completely rewrite Edison's response.

Students provided their consent for 199 out of 435 questions for which we have asked for consent. TAs activated Edison for 110 questions (after removing miscellaneous activations). There were 15 unique users (i.e., TAs), with the majority of the calls (~65%) made by three experienced TAs; all the TAs have previously taken the course (Data 100) and were either enrolled in or had completed a pedagogy course at that time.

3.2 Edison's Pipeline

Edison utilizes historical Q&A pairs and course documents to answer new student questions with RAG. The system is also augmented with a few additional features (the details of the base pipeline are presented in [31]). Images attached as part of the context are parsed using OCR and appended to each question. Based on student-tagged metadata, incoming questions about assignments are also augmented with the solutions from the corresponding assignment. Syllabus text and course notes were used to create *logistics* and *conceptual* vector databases, respectively. As a preprocessing step, an incoming question is first passed to GPT-4 to determine its category - *logistics*, *assignment*, or *conceptual*. Within each of these categories, only appropriate vector databases were queried, and different category-based prompts, curated with in-category hand-engineered few-shot examples, were used to generate an answer. Finally, answers generated for *assignment* questions were passed to an additional GPT-4 call to make the answer concise and ensure it does not contain solution-revealing hints. Based on qualitative observations, the additional features - OCR, category-based conditional prompting, few-shot examples, and additional LLM solution filtering - improved the performance of the base pipeline [31]; a thorough evaluation of the adapted system is left for future work.

[1] For additional information, please refer to the course website: https://ds100.org/sp24/

[2] https://edstem.org/

[3] The reason for not deploying for an entire semester was due in part to the complexity of the deployment, which delayed the study.

Table 1: TAs evaluated Edison's responses according to factuality, relevance, and style.

Evaluation metric	Description
Factuality	Evaluates the correctness of the information provided in the response.
Relevance	Evaluates the degree to which the response is pertinent or related to the given student question and course.
Style	Evaluates the degree to which the coherence, length, and the use of solutions, hints, and examples are appropriate for the given student question.

Table 2: TA edit intent categories (hints/sols, style, content, relevance), category descriptions, and representative examples (truncated for brevity). The first and last examples are deletions; the second and third examples are modifications.

Category	Description	Example TA Edit
Hints/sols	Edits removing hints or solutions or adding hints for additional guidance.	[deleted] Ensure you complete this aggregation step to convert the binary...
Style	Edits addressing tone, style, and verbosity of the response.	[original] Let's clarify: The singular values in S represent the amount of variance... [modified] The singular values in S represent the amount of variance...
Content	Edits improving the factuality of the response within the course context.	[original] Eigenvectors give us the directions of the axes of maximum variance. [modified] Each eigenvector of the covariance matrix gives a direction in the feature...
Relevance	Edits improving the relevance of the response within the course context.	[deleted] ...using tf-idf scores or considering bigrams/trigrams could provide more...

3.3 Data Processing

There were a total of 110 calls made to Edison; 65 were for assignment, 41 for conceptual, and 4 for logistical questions. Due to the limited sample size, we did not use the responses to logistical questions in this paper. The initial Edison response, TA ratings (see subsection 3.1), and the final answers after TA edits were recorded. We categorize TA edits based on their intent as well as based on their mechanistic implementation.

First, sentence-level edits between the initial Edison response and the edited TA answer were extracted[4]. The three co-authors of this study, two of which have extensive experience as TAs for large-scale CS or data science courses, reviewed a sample of Edison responses alongside corresponding TA edits and tagged the *intent* of each edit. Afterward, the annotators discussed and arrived at the final four categories — hints and solutions, style, content, and relevance. Descriptions and representative examples of each intent category are shown in Table 2. All three annotators labeled a subset of 10 edited responses with a total of 35 edits; the initial inter-rater reliability rate was 71%. After discussing and resolving inconsistencies, the final agreement rate was 85%. Then, the rest of the data was divided equally among the researchers and labeled according to the discussed criteria.

We also consider the *mechanistic* implementation of the edits (different from the intent categories described above) — additions, deletions, and modifications. Mechanistic categories are defined on sentence level (e.g., modification is when a sentence is changed, and addition is when a new sentence is added). Furthermore, the number of edited units on a sentence/paragraph level was counted (e.g., deleting multiple sequential sentences was considered one deleted unit).

[4]We used difflib python library to automatically extract sentence-level differences between two strings.

In addition to sentence-level analysis, we looked at edits on the word level as well. For modification type edits, word-level Levenshtein edit distance (referred to hereafter as edit distance) was calculated [22]. For deletions and additions, we used the total number of words deleted or added, respectively.

We also analyze the difference in the effort it takes for TAs to answer student questions with and without using Edison; with Edison, they start with the initial response and either use it verbatim, modify it, or rewrite it. To quantify TA effort without using Edison, we use TA answers to questions where Edison was not triggered. We classify responses into three categories: Edison-only, where the TA uses the response without modification; TA+Edison, where the TA edits or rewrites the response; and TA-only, where Edison is not triggered. We filtered assignment-related and conceptual questions posted during the deployment timeframe to minimize the potential biases between the two datasets. The number of words in the TA-only answer was used to approximate their effort, factored as an addition-type edit. To calculate the TA effort it takes to produce TA+Edison answers, we use three proxies:

- **Naive cost:** We weigh additions, deletions, and modifications equally, and the total effort is the summation of the corresponding edit distances on a word level: `a+m+d`, where `a`, `m`, and `d` are the edit distances of additions, modifications, and deletions, respectively.
- **Adjusted weights cost:** Removing existing text takes less cognitive strain and time than writing new text [20]. As deletions take less effort, we weigh deletions as half the effort of additions and modifications: `a+m+d/2`.
- **Modified units cost:** We also define cost in terms of modification units. In this case, the total effort is calculated as the edit distance of additions and modifications plus the number of deleted units (e.g., sentences, paragraphs): `a+m+units deleted`. The rationale for this metric is that TAs often remove a substantial portion of the response to make it less verbose, and

the cost of removing sequential sentences and words can be counted as one unit.

4 Results

4.1 RQ1: TA Evaluation of Edison

Figure 1 presents the results of the TA evaluations of Edison-only responses [5].

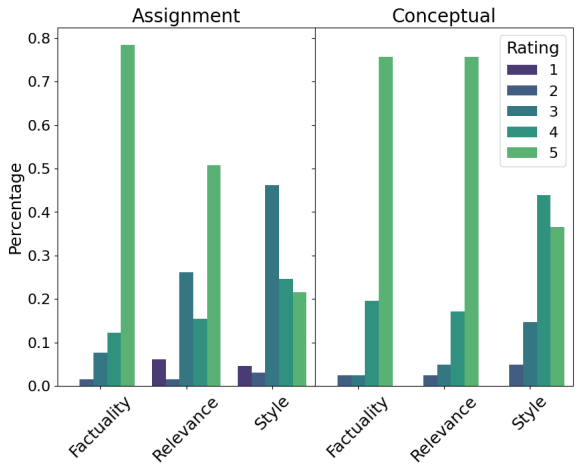


Figure 1: Assignment (left) and conceptual (right) TA evaluations according to factuality, relevance, and style shown as a distribution over a 1-5 scale. The y-axis shows the proportion of each rating.

Factuality: The responses are rated relatively highly (5) on factuality for both assignment (51/65) and conceptual (31/41) questions. The main failure cases for assignment questions occurred when the model could not identify the main problem with the student's approach.

Relevance: For assignment questions, the responses are rated lower (43/65 rated 4 or 5) compared to conceptual questions (38/41). The main failure cases involve responses where the model included additional irrelevant suggestions or information not pertinent to the class context. An example of such a response is as follows:

> The message you're seeing means that your logistic regression model couldn't find the optimal solution within the default number of iterations. Here are a couple of hints: 1. Increase the Number of Iterations... 2. Scale Your Data...

The second suggestion is out of scope in the course context, so the TA rated this response as a three for relevance. In contrast, here is a response that was rated highly (5) for relevance.

> Absolutely, I'd be glad to clarify the difference between A and B in the silhouette score formula for you...In the silhouette score formula, $S = \frac{B-A}{\max(A,B)}$, A represents the average distance to other points within the same cluster as the data point X. Essentially, it measures...B is the average distance from X to the points in the closest cluster that X is not a part of. This measures how well-separated X is from the nearest cluster...

The response above is grounded and relevant within the context of the course and the student's question; the response is based on the relevant retrieved content from the course textbook.

Style: For both assignment (30/65 rated 4 or 5) and conceptual (37/65 rated 4 or 5) questions, Edison's responses are rated lower on style compared to factuality and relevance. Verbosity and excessive hints are the main issues. Below is an example of a response that gave away a significant part of the solution and was rated as one for style.

> Ah, I see where the confusion might be coming from...Try adjusting your `WHERE` clause to something like this...<solution code>...This adjustment makes sure that both conditions...

In contrast, below is an example of a response that was rated as a five. It identifies the most probable issue based on the student's code and error message and provides appropriate guidance without giving away too much information in the hint.

> Consider revising how you construct your DataFrame, ensuring contains_body and contains_head are treated as data columns. This will establish the correct structure for computing correlations...After adjusting the DataFrame, proceed with calculating the correlation and creating the heatmap with seaborn as intended...

4.2 RQ2: Characteristics of TA Edits

4.2.1 Length Analysis. Figure 2 shows the lengths of TA-only, Edison-only, and TA+Edison responses. The final response after the TA edits is shorter on average compared to the initial response generated by Edison. Furthermore, on average, TAs shorten answers to assignment questions more than conceptual questions, as seen by the gap between the solid and the dashed lines in the left (red) and right (blue) panels. The TA-only responses tend to be shorter than the TA+Edison responses for both assignment and conceptual questions (gap between solid black and colored curves). When we look at examples, TA+Edison responses often include supporting examples with step-by-step explanations, which makes them longer. We will investigate why TA responses are shorter and the implications of this in future work.

4.2.2 Edit Category and Type Distribution. Out of 106 Edison-generated responses, TAs edited 61 (57.5%), directly used 28 as-is (26.4%), and rewrote 17 from scratch (16.0%). While rewrites may require more effort than TA-only responses, direct use or edits in 84% of cases save significant time. These statistics broken down by the type of the question are presented in Table 3. On average, there are more edits to responses to assignment (89.39 words) than conceptual questions (65.11 words).

We observe that Edison performs better on conceptual questions, as indicated by the proportion of responses where the TA either directly used (46.3%) or edited (46.3%) Edison's output. In contrast, for assignment questions, TAs directly used Edison's response in only 13.8% of cases, while 64.6% required edits. Additionally, TAs

[5]We do not display average results as the ratings are on the Likert scale.

Table 3: TA edit statistics categorized by question type (assignment and conceptual). The first three columns show the number of Edison's responses that TAs edited, directly used as-is, or rewrote from scratch. The last columns show the average word edit distance by mechanistic edit type (addition, deletion, and modification) and edit intent category (hints/solutions, style, content, and relevance) along with the overall statistics. The numbers in the parentheses are the proportions of the total edit distance within the edit categories (e.g. the proportion of addition edit distance of the total edit distance for style edits).

	# Edited	# Used as-is	# Rewrites	Category	Add distance	Del distance	Mod distance	Total distance
Assignment	42 (64.6%)	9 (13.8%)	14 (21.5%)	Hints/sols	2.39 (16.9%)	10.67 (75.6%)	1.06 (7.5%)	14.12
				Style	0.65 (1.5%)	39.65 (89.8%)	3.86 (8.7%)	44.16
				Content	4.47 (57.2%)	2.96 (37.9%)	0.39 (5.0%)	7.82
				Relevance	0.53 (2.3%)	22.76 (97.7%)	0.00 (0.0%)	23.29
				Total	**8.04 (9.0%)**	**76.04 (85.1%)**	**5.31 (5.9%)**	**89.39**
Content	19 (46.3%)	19 (46.3%)	3 (7.3%)	Hints/sol	0.00 (0.0%)	0.00 (0.0%)	0.00 (0.0%)	0.00
				Style	4.61 (8.4%)	44.34 (81.2%)	5.58 (10.2%)	54.63
				Content	5.42 (60.1%)	2.32 (25.1%)	1.16 (13.0%)	8.89
				Relevance	0.84 (53.2%)	0.00 (0.0%)	0.74 (46.8%)	1.58
				Total	**10.86 (16.7%)**	**46.66 (71.7%)**	**7.58 (11.6%)**	**65.11**

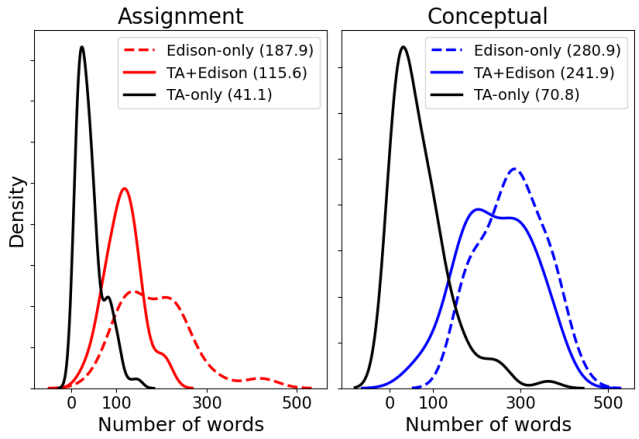


Figure 2: Assignment (left) and conceptual (right) response lengths of TA-only, Edison-only, and TA+Edison (Edison response after TA edits). The numbers in the parentheses are the average response lengths.

rewrote 21.5% of responses for assignment questions from scratch, compared to just 7.3% for conceptual questions.

Assignment questions: A large proportion of edits address style-related problems (e.g., verbosity), with most being deletions (89.8%). A total of 97.7% of edits attributed to improving response relevance are deletions. If the response contained irrelevant sections, the TA typically removed those sections; however, if the response was overall irrelevant, the TA rewrote the answer from scratch. Edits addressing hints and solutions were also mainly deletions (75.6%), as the TAs mainly removed unnecessary hints; for some cases, the TA decided to add hints to Edison's response (16.9%).

Conceptual questions: For conceptual questions, the main edits were again style-related deletions. However, more edits were improving the factuality of the response (13.65%) compared to those of assignment questions (8.75%). Fewer edits addressed the relevance of the responses (2.43%).

4.3 RQ3: TA Rewrite Analysis

To analyze TAs' decisions to use or rewrite Edison's responses, we report the pairwise correlations between the three evaluation metrics and the rewrite indicator variable (1 if it was a rewrite, 0 otherwise) in Figure 3. The coefficients are -0.06, -0.5, and -0.004 for factuality, relevance, and style metrics, respectively. Only the metric corresponding to relevance is statistically significant, indicating that relevance is most correlated to the TA's decision to completely rewrite the answer. While most edits focus on improving the response's style, style is not strongly correlated with the decision to rewrite. This finding highlights opportunities for tuning LLMs to better align with the stylistic needs of educational settings.



Figure 3: Pairwise correlations between evaluation metrics (factuality, relevance, and style) and the rewrite indicator variable.

Table 4: Estimated effort required to write answers with (TA+Edison and Edison-only) and without (TA-only) Edison, using three methods (see subsection 3.3). Highlighted numbers in each effort column indicate a statistically significant lower effort compared to the other; if no numbers are highlighted, the difference is not statistically significant. The number of asterisks denotes the significance level of the difference. Numbers in parentheses represent standard errors.

		Add dist	Del dist	Mod dist	Naive Cost Effort (a+m+d)	Adjusted Effort (a+m+d/2)	Modified Unit Effort (a+m+del units)
Assignment	TA-only	40.69 (1.77)	-	-	**40.69*** (1.77)**	40.69 (1.77)	40.69 (1.77)
	With Edison	15.11 (3.15)	59.66 (10.75)	4.17 (1.16)	78.94 (11.58)	49.11 (6.55)	19.28 (3.09) + 1.68 (0.25)
Conceptual	TA-only	70.46 (6.7)	-	70.46 (6.7)	70.46 (6.7)	70.46 (6.7)	70.46 (6.7)
	With Edison	18.41 (6.73)	43.24 (10.29)	7.02 (2.31)	68.68 (15.43)	**47.06* (11.09)**	25.44 (7.8) + 0.88 (0.22)
Overall	TA-only	48.47 (2.28)	-	-	**48.47** (2.28)**	48.47 (2.28)	48.47 (2.28)
	With Edison	16.39 (3.23)	53.31 (7.7)	5.27 (1.14)	74.97 (9.24)	48.32 (5.85)	21.66 (3.55) + 1.37 (0.18)

4.4 RQ4: Efficiency Gain Analysis

To analyze the comparative effort it takes TAs to answer questions with and without the use of Edison, we use multiple methods for calculating effort (see subsection 3.3); the results are shown in Table 4.

- **Naive cost** (`a+m+d`): The effort to respond without Edison (TA-only) is lower for assignment questions ($p < 0.05$), but not statistically significant for conceptual questions.
- **Adjusted weights cost** (`a+m+d/2`): The effort to respond with Edison is lower for conceptual questions ($p < 0.05$), but not statistically significant for assignment questions.
- **Modified units cost** (`a+m+units deleted`): The estimated effort is 48.47 for the TA and 21.66 + 1.37 units deleted for TA+Edison.

It should be mentioned that we do not account for the cognitive effort required by TAs to mentally process the question and formulate a response. Edison can potentially reduce this cognitive load by providing accurate responses for quick edits and verification.

5 Limitations & Threats to Validity

One limitation of our work is evaluating our method in just one course. The nuances of Edison's performance on other courses should be further investigated. Furthermore, while evaluation by expert TAs offers high-quality insights, it also limits the scope of our analysis to a small number of samples. If we were to use an LLM-based judge (appropriately trained) we could potentially scale this evaluation further. Beyond considerations of scope and scale, several external factors may influence the validity of our results. One such factor is the extent to which students independently use LLMs[6]. Another factor is the fluctuation in the volume of student forum posts over time, which could impact the consistency and representativeness of the data collected during the study. Lastly, the efficiency analysis does not account for the cognitive load needed to write an answer when not using Edison or reading through and editing the initial answer when using Edison.

6 Conclusion

This paper analyzes how expert TAs evaluate responses from an LLM educational assistant. We further explore how TA edits to LLM-generated responses provide insights for enhancing their quality to better align with teaching goals. Our work features several key insights: (1) Edison, while highly factual, can be improved to provide more relevant responses to assignment questions and less verbose responses overall; (2) On average, Edison generates longer answers than TAs. While this is preferable for conceptual questions, shorter answers are better for assignment questions; (3) We find that most edits by TAs are made to improve response style; (4) Only 16% of responses were rewritten, and the relevance of the response is the most strongly correlated factor to the TA's decision to rewrite the response; and (5) Edison answers save TAs time when answering content questions.

These findings offer valuable insights into the differences in instructional effectiveness between LLM and high-quality TA responses, informing the efficacy and future improvements of educational LLM assistants in real-world deployment.

[6]Technically prohibited in the course syllabus for use on assignments.

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